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RESULTS

Results

01 · AVERAGE VARIANCE EXTRACTED (AVE)

convergent and discriminant validity

Table 1. Convergent validity

Construct	AVE	CR	ω	α
esg	.60	.86	.86	.86
norm	.65	.88	.88	.88
intent	.65	.84	.84	.84
attitude	.65	.88	.88	.88
service_quality	.62	.86	.86	.86

Table 2. Discriminant validity (Fornell–Larcker)

	esg	norm	intent	attitude	service_quality
esg	.78				
norm	–.11	.81			
intent	.37	.26	.80		
attitude	–.02	–.03	.32	.81	
service_quality	.00	–.14	.23	.16	.79

Fornell–Larcker: Diagonal = $\sqrt{\text{AVE}}$ (bold); off-diagonal = inter-construct latent correlations.

Table 3. Discriminant validity (HTMT)

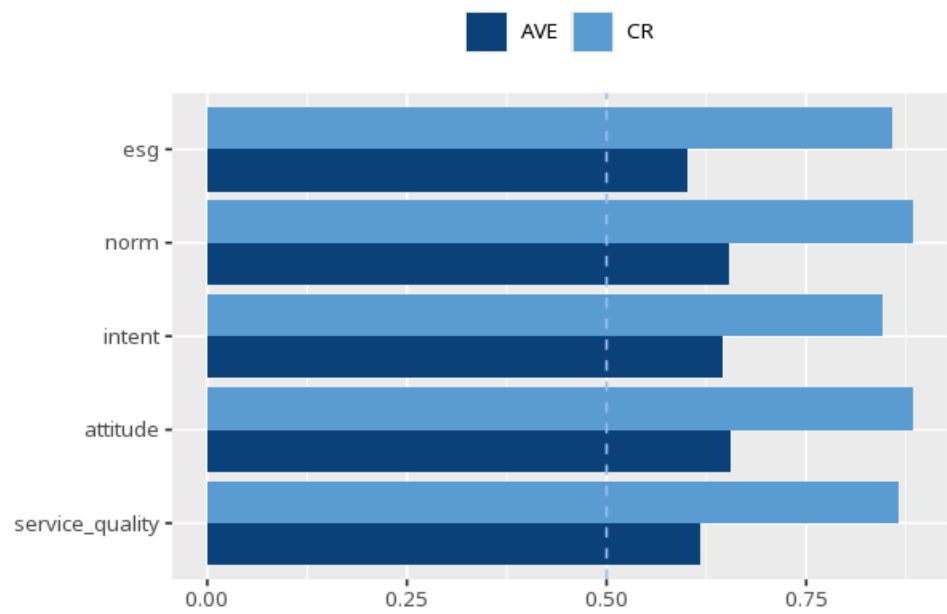
	esg	norm	intent	attitude	service_quality
esg					
norm	.10				
intent	.37	.26			
attitude	.03	.03	.31		
service_quality	.05	.13	.23	.14	

Scope: When only 1 construct is defined, Tables 2 and 3 are suppressed (discriminant validity requires ≥ 2 constructs). Applies to reflective constructs only.

Cutoffs: AVE $\geq .50$ indicates adequate convergent validity, computed from the CFA loadings (Fornell & Larcker, 1981); CR $\geq .70$ is acceptable (Nunnally, 1978; Bagozzi & Yi, 1988); do not cite CR $\geq .70$ to Fornell & Larcker; HTMT $< .85$ indicates discriminant validity for conceptually distinct constructs (Henseler, Ringle & Sarstedt, 2015).

Reliability: ω (McDonald's) is the preferred reliability coefficient — model-based, not assuming tau-equivalence (McNeish, 2018); α is retained as a secondary/legacy column.

Figure. Validity overview



Understanding the numbers

AVE. The average share of a construct's item variance that it actually captures; $\geq .50$ (Fornell & Larcker, 1981) means the items converge well on their construct. Each row is one construct's own AVE; a construct below .50 may still be acceptable when its CR is above .60 (the Fornell & Larcker caveat), but treat that as a caution, not a lenient pass.

CR. Composite reliability - how reliably a construct's indicators measure it as a group; $\geq .70$ is acceptable (Nunnally, 1978; Bagozzi & Yi, 1988). Each row is one construct's own CR; a value below .70 flags a construct whose indicators may not hang together well.

ω (omega). The preferred headline reliability coefficient per construct - model-based, not assuming tau-equivalence (McNeish, 2018). Compare each construct's ω against its α in the same row - a large gap suggests the items' loadings are not equal.

α (alpha). Cronbach's alpha, shown per construct as a secondary/legacy coefficient alongside ω . Each construct's α is shown for reference alongside its own ω , the preferred coefficient.

How to read this test

AVE (Average Variance Extracted) is the average share of item variance a construct captures; $\geq .50$ means items converge on their construct (Fornell & Larcker, 1981). CR (Composite Reliability) $\geq .70$ indicates acceptable construct reliability. ω (McDonald's omega) is the preferred headline reliability coefficient; α (Cronbach's) is shown as a secondary/legacy coefficient. For discriminant validity, Fornell–Larcker (Table 2): each construct's $\sqrt{\text{AVE}}$ (bold diagonal) should exceed its

correlations with other constructs. HTMT (Table 3): values < .85 support discriminant validity for conceptually distinct constructs — use HTMT as the primary criterion and Fornell–Larcker as a legacy check. AVE < .50 may still be acceptable when CR > .60 (F&L caveat), but treat this as a caution, not a lenient pass. All cutoffs are heuristics, not pass/fail gates. Applies to reflective constructs only.

APA template: Convergent validity was supported, all AVE $\geq$.50 and CR $\geq$.70; discriminant validity held, all HTMT < .85.

Statistical basis: AVE (average variance extracted) convergent validity, $\geq$.50 (Fornell, C., Larcker, D. F. (1981). "Evaluating structural equation models with unobservable variables and measurement error." *Journal of Marketing Research*, 18(1), 39-50.); CR (composite reliability) $\geq$.70 acceptable (Nunnally, J. C. (1978). *Psychometric Theory* (2nd ed.). McGraw-Hill.); CR $\geq$.70 threshold, independent corroboration (Bagozzi, R. P., Yi, Y. (1988). "On the evaluation of structural equation models." *Journal of the Academy of Marketing Science*, 16(1), 74-94.); HTMT < .85 discriminant validity (primary criterion) (Henseler, J., Ringle, C. M., Sarstedt, M. (2015). "A new criterion for assessing discriminant validity in variance-based structural equation modeling." *Journal of the Academy of Marketing Science*, 43(1), 115-135.); Reliability preference (ω over α) (McNeish, D. (2018). "Thanks coefficient alpha, we'll take it from here." *Psychological Methods*, 23(3), 412-433.). Full references in the exported CITATIONS.txt.

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